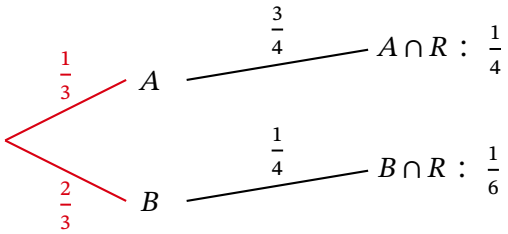
Reversing a conditional probability

High school · probability · Lesson and method · 07

Prerequisites. Probability trees, intersections and conditional probability.

Key point · Ideas and method. $P(A|B)$ is the probability of A given that B occurred. If $P(B) > 0$, $P(A|B) = \frac{P(A \cap B)}{P(B)}$. The information B restricts the sample space.
$P(A \cap B) = P(A)P(B|A)$. If A and its complement form the first two branches of a tree, $P(B) = P(A)P(B|A) + P(\text{not } A)P(B|\text{not } A)$.
Bayes' formula follows from these two identities. $P(A|B)$ generally differs from $P(B|A)$: the denominators refer to different populations.

Example 1. Choose urn A with probability $\frac{1}{3}$ and B with probability $\frac{2}{3}$. A contains 3 red and 1 blue ball; B contains 1 red and 3 blue balls. $P(\text{red}) = (\frac{1}{3})(\frac{3}{4}) + (\frac{2}{3})(\frac{1}{4}) = \frac{5}{12}$. Given a red ball, $P(A|\text{red}) = \frac{\frac{1}{4}}{\frac{5}{12}} = \frac{3}{5}$.



Only branches leading to a red ball are shown.

Activity 1 · Explore. Represent the example with 120 hypothetical trials matching the proportions exactly. Count red balls from each urn.